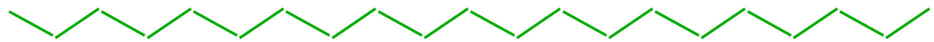
Number average molecular weight

$M_1 = 100$ 

$N_1 = 1$

$M_2 = 10$ 

$N_2 = 1$

number fraction : $n_i = \dfrac{\text{number of type } i}{\text{total number}}$

$\rightarrow$ $n_1 = 1/2$; $n_2 = 1/2$

$$\bar{M}_n = \sum n_i M_i = \frac{1}{2} \cdot 100 + \frac{1}{2} \cdot 10 = 55$$

$$\bar{\boldsymbol{M}}_{\boldsymbol{n}} = \frac{\sum \boldsymbol{n}_{\boldsymbol{i}} \boldsymbol{M}_{\boldsymbol{i}}}{\sum \boldsymbol{n}_{\boldsymbol{i}}}$$

$$n_i = \frac{N_i}{\Sigma N} \qquad \therefore\ n_1 = \frac{1}{2},\ n_2 = \frac{1}{2}$$

$$w_i = \frac{N_i M_i}{\Sigma N_i M_i} \qquad \therefore\ w_1 = \frac{10}{11},\ w_2 = \frac{1}{11}$$

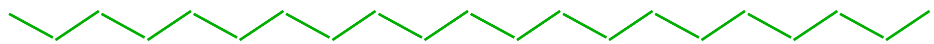
$$z_i = \frac{N_i M_i^2}{\Sigma N_i M_i^2} \qquad \therefore\ z_1 = \frac{100}{101},\ z_2 = \frac{1}{101}$$

$$\therefore\ M_n = \Sigma n_i M_i = 55$$

$$M_w = \Sigma w_i M_i = \frac{1010}{11} = 91.82$$

$$M_z = \Sigma z_i M_i = \frac{10010}{101} = 99.12$$

Weight average molecular weight

$M_1 = 100$ 

$N_1 = 1$

$M_2 = 10$ 

$N_2 = 1$

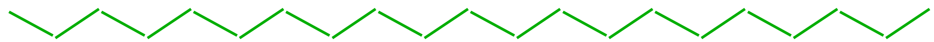
weight fraction : $w_i = \dfrac{\text{weight of type } i}{\text{total weight}}$

$\rightarrow$ $w_1 = 10/11$; $w_2 = 1/11$

$$\bar{M}_w = \sum w_i M_i = \frac{10}{11} \cdot 100 + \frac{1}{11} \cdot 10 = 91.8$$

$$\bar{M}_w = \frac{\sum n_i M_i^2}{\sum n_i M_i}$$

Z-average molecular weight

$M_1 = 100$ 
$N_1 = 1$

$M_2 = 10$ 
$N_2 = 1$

Z-fraction : $z_i = \frac{N_i M_i^2}{\sum N_i M_i^2}$

$\rightarrow$ $z_1 = 100/101$; $z_2 = 1/101$

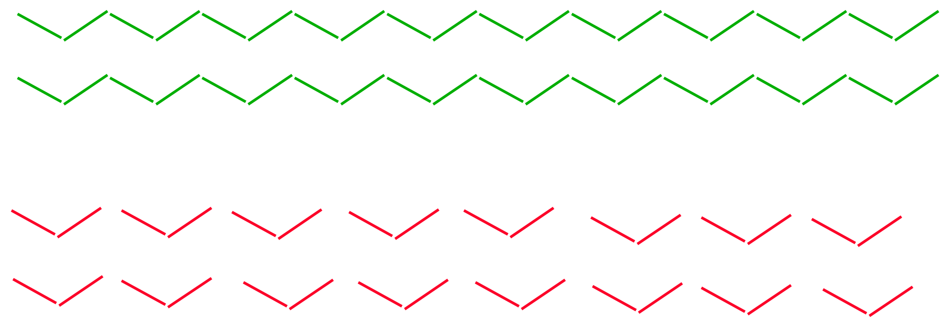
$$\bar{M}_z = \sum z_i M_i = \frac{100}{101} \cdot 100 + \frac{1}{101} \cdot 10 = 99.1$$

$$\bar{M}_z = \frac{\sum n_i M_i^3}{\sum n_i M_i^2}$$

Number average molecular weight

$M_1 = 100$
$N_1 = 2$

$M_2 = 10$
$N_2 = 16$



number fraction : $n_i = \dfrac{\text{number of type } i}{\text{total number}}$

$\rightarrow$ $n_1 = 2/18$; $n_2 = 16/18$

$$\bar{M}_n = \sum n_i \cdot M_i = \frac{2}{18} \cdot 100 + \frac{16}{18} \cdot 10 = 20$$

$$\bar{M}_n = \frac{\sum n_i M_i}{\sum n_i}$$

$$\sum N_i = 18$$

$$\sum N_i M_i = 360$$

$$\sum N_i M_i^2 = 21600$$

$$\sum N_i M_i^3 = 2016000$$

$$\therefore \bar{M}_n = \frac{\sum N_i M_i}{\sum N_i} = 20$$

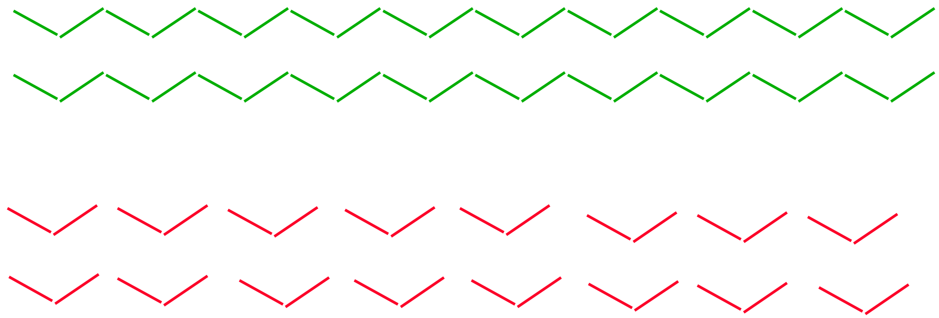
$$\bar{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i} = 60$$

$$\bar{M}_z = \frac{\sum N_i M_i^3}{\sum N_i M_i^2} = 93.\dot{3}$$

Weight average molecular weight

$M_1 = 100$
$N_1 = 2$

$M_2 = 10$
$N_2 = 16$



weight fraction : $w_i = \dfrac{\text{weight of type } i}{\text{total weight}}$

$\rightarrow$ $w_1 = 200/360$; $w_2 = 160/360$

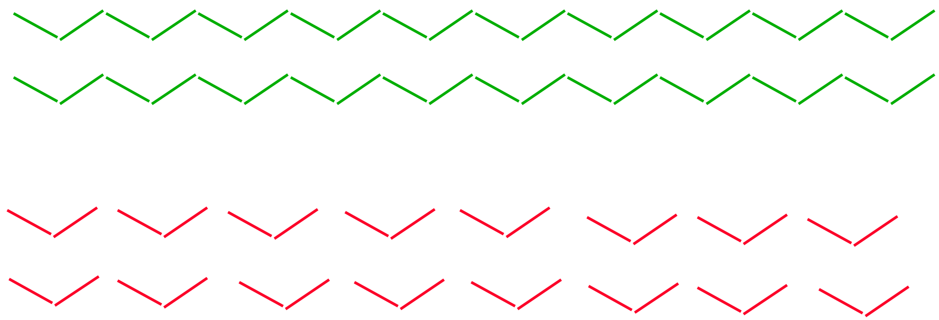
$$\bar{M}_w = \sum w_i \cdot M_i = \frac{200}{360} \cdot 100 + \frac{160}{360} \cdot 10 = 60$$

$$\bar{M}_w = \frac{\sum n_i M_i^2}{\sum n_i M_i}$$

Z-average molecular weight

$M_1 = 100$
$N_1 = 2$

$M_2 = 10$
$N_2 = 16$



Z-fraction : $z_i = \frac{N_i M_i^2}{\sum N_i M_i^2}$

$\rightarrow \quad z_1 = 200/216 \,;\, z_2 = 16/216$

$$\bar{M}_z = \sum z_i \cdot M_i = \frac{200}{216} \cdot 100 + \frac{16}{216} \cdot 10 = 93.3$$

$$\bar{M}_z = \frac{\sum n_i M_i^3}{\sum n_i M_i^2}$$

Example of averages calculations

9 molecules of M = 30,000 and 5 molecules of M = 50,000

- Number-average molar mass:

$$\overline{M}_n = \frac{\sum_{i=1}^{\infty} M_i N_i}{\sum_{i=1}^{\infty} N_i} = \frac{(9\times 30{,}000)+(5\times 50{,}000)}{(9+5)} = 37{,}000$$

- Weight-average molar mass:

$$\overline{M}_w = \frac{\sum_{i=1}^{\infty} w_i M_i}{\sum_{i=1}^{\infty} w_i} = \frac{\sum_{i=1}^{\infty} N_i M_i^{\,2}}{\sum_{i=1}^{\infty} N_i M_i} = \frac{9(30{,}000)^2 + 5(50{,}000)^2}{9(30{,}000)+5(50{,}000)} = 40{,}000$$

- Z-average molar mass:

$$\overline{M}_z = \frac{\sum_{i=1}^{\infty} N_i M_i^3}{\sum_{i=1}^{\infty} N_i M_i^2} = \frac{\sum_{i=1}^{\infty} w_i M_i^{\,2}}{\sum_{i=1}^{\infty} w_i M_i} = \frac{9(30{,}000)^3 + 5(50{,}000)^3}{9(30{,}000)^2 + 5(50{,}000)^2} = 42{,}136$$

$$\sum N_i = 14$$

$$\sum N_i M_i = 520000$$

$$\sum N_i M_i^2 = 2.06 \times 10^{10}$$

$$\sum N_i M_i^3 = 8.68 \times 10^{14}$$

$$\therefore \bar{M}_n = \frac{\sum N_i M_i}{\sum N_i} = 37142.86$$

$$\bar{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i} = 39615.38$$

$$\bar{M}_z = \frac{\sum N_i M_i^3}{\sum N_i M_i^2} = 42135.92$$